Vector fields

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Given an open set $U \subseteq \mathbb{R}^n$, a **vector field** X over U is an object of type

$$X :: (p : U) \to T_p(U)$$

So a given vector field associates, for every point $p \in U$, a vector with its tail at p.

Recall the isomorphism $T_p(U) \cong \mathbb{R}^n \cong \mathcal{D}\big(C_p^\infty\big)$.

So, we can take basis for $T_p(U)$ as $\{e_1, ..., e_n\}$ (which is actually a basis for $\mathbb{R}^n$), and $\left\{ \frac{\partial}{\partial x_1}\Big|_p, ..., \frac{\partial}{\partial x_n}\Big|_p \right\}$ for $\mathcal{D}\big(C_p^\infty\big)$. So, we can write the 3-dimensional vector field $X_p := [p_x \;\; 0 \;\; -p_y^2]^T$ as

$$(p_x, p_y, p_z) \mapsto p_x \frac{\partial}{\partial x}\bigg|_p - p_y^2 \frac{\partial}{\partial z}\bigg|_p$$

Given a vector field X, for example $X_p := p_x \frac{\partial}{\partial x}\big|_p - p_y^2 \frac{\partial}{\partial z}\big|_p$, and $f \in C^\infty(U)$, we can define

$$(Xf)_p = p_x \frac{\partial f}{\partial x}\bigg|_p - p_y^2 \frac{\partial f}{\partial z}\bigg|_p$$

Observe that Xf is a function from U to $\mathbb{R}$. If X is a C^∞-smooth vector field, then Xf will be in $C^\infty(U)$, so let us first define what a C^∞-smooth vector field means.

Recall that the isomorphism $T_p(U) \xrightarrow{\sim} \mathcal{D}\big(C_p^\infty\big)$ converts a vector v based at p into the action of taking the directional derivative of a given function at p along v.

So given a vector field X, the above isomorphism converts X_p from being a vector in $T_p(U)$ to being the action of taking directional derivative at p along X_p.

Given a vector field X on $U \subseteq \mathbb{R}^n$, we can define the "component functions" of X as $\{\pi_i \circ X\}_{1 \le i \le n}$, where π_i projects $T_p(U)$ onto the basis vector e_i (or equivalently, projects $\mathcal{D}\big(C_p^\infty\big)$ onto the basis vector $\frac{\partial}{\partial x_i}\Big|_p$).

A vector field X is called $\boldsymbol{C^\infty}$**-smooth** if all of the component functions are smooth. This implies that every smooth vector field X can be expressed as

$$X_p = \sum_{i=1}^{n} f_i(p) \frac{\partial}{\partial x_i}\bigg|_p, \text{for some } f_i \in C^\infty(U).$$

We denote by $\mathcal{X}(U)$ **the space of all smooth vector fields over** U. The above statement therefore becomes that

$$\mathcal{X}(U) \cong \bigoplus_{i=1}^{n} \left[C^\infty(U) \frac{\partial}{\partial x_i} \right]$$

This makes $\mathcal{X}(U)$ into a $C^\infty(U)$-module!

From now onwards, we shall assume that our vector fields are C^∞-smooth unless otherwise stated.

Let us revisit $f \mapsto Xf$, where $X \in \mathcal{X}(U), f \in C^\infty(U)$.

Thinking of $\mathcal{X}(U)$ as $(p : U) \to \mathcal{D}\left(C_p^\infty\right)$, the above map simply becomes

$$Xf := \left[(p : U) \mapsto X_p(f) \right]$$

Since X is C^∞-smooth, so let $X_p = \sum a_i(p) \frac{\partial}{\partial x_i}\Big|_p$ for some $a_i \in C^\infty(U)$. Therefore,

$$(Xf)_p = \sum a_i(p) \frac{\partial f(p)}{\partial x_i} \in C^\infty(U)$$

Therefore, observe that $\Phi_X := f \mapsto Xf$ is a map from $C^\infty(U) \to C^\infty(U)$ which has the following properties:

- It is $\mathbb{R}$-linear
- It satisfies the Leibniz rule, $\Phi_X(fg) = f\Phi_X(g) + g\Phi_X(f)$, because

$$\begin{aligned}
(X(fg))_p &= \sum a_i(p) \frac{\partial (fg)(p)}{\partial x_i} \\
&= \sum a_i(p) \left[f(p) \frac{\partial g(p)}{\partial x_i} + g(p) \frac{\partial f(p)}{\partial x_i} \right] \\
&= f(p)(Xg)_p + g(p)(Xf)_p
\end{aligned}$$

A $C^\infty(U)$ to $C^\infty(U)$ function which is $\mathbb{R}$-linear and satisfies the Leibniz rule is called a **derivation of** $C^\infty(U)$. We denote the set of all derivations of $C^\infty(U)$ as $\mathrm{Der}(C^\infty(U), C^\infty(U))$.

Note that this is *different* from a point derivation of C_p^∞ at p, which we have denoted as $\mathcal{D}\left(C_p^\infty\right)$ previously.

What does Xf "actually" represent?

View #1

> X assigns to every point $p \in U$ the vector $X_p \in T_p(U)$. But, we have seen in the previous part that every vector corresponds to the point derivation given by taking the directional derivative at that point.
>
> Therefore, we can think of X_p as the action of taking the directional derivative of a given function at the point p along the vector X_p.
>
> Given a function $f \in C^\infty(U)$, $(Xf)_p$ is the directional derivative of f along X_p.

View #2

Given the smooth 2-dimensional vector field $X = \begin{bmatrix} -y & x^2 \end{bmatrix}^T$, we can also write it as $X = -y\frac{\partial}{\partial x} + x^2\frac{\partial}{\partial y}$.

Now, given $f \in C^\infty(U)$, we have defined $Xf \in C^\infty(U)$ as

$$(Xf)_p := -p_y \left.\frac{\partial f}{\partial x}\right|_p + p_x^2 \left.\frac{\partial f}{\partial y}\right|_p$$

So, at a point $p \in U$, $(Xf)_p$ is the dot product of the vector field at that point with the gradient of f at that point. Basically, it expressed how "aligned" the vector field is with the direction of steepest ascent of f.

In the next section, we shall see the definition of grad and others.

Let us discuss about grad, curl and div. U_n denotes an arbitrary open subset of $\mathbb{R}^n$.

Gradient (grad)

Given a scalar function $f \in C^\infty(U_n)$, we can define a vector field

$$\text{grad}(f) \in \mathcal{X}(U_n)$$

$$\text{grad}(f)_p := \begin{bmatrix} \frac{\partial f(p)}{\partial x_1} & \dots & \frac{\partial f(p)}{\partial x_n} \end{bmatrix}^T$$

$\text{grad}(f)_p$ measures the direction and magnitude of the steepest ascent of f at the point p.

Curl (curl)

Curl is conventionally defined over 2-dimensional and 3-dimensional vector fields. We shall denote them respectively by curl_2 and curl_3.

curl_3 takes in a 3-dimensional vector field and produces another 3-dimensional vector field. Given a vector field $X = f_x\frac{\partial}{\partial x} + f_y\frac{\partial}{\partial y} + f_z\frac{\partial}{\partial z} \in \mathcal{X}(U_3)$, we define

$$\text{curl}_3(X) \in \mathcal{X}(U_3)$$

$$\text{curl}_3(X)_p = \det\left(\begin{bmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ \frac{\partial}{\partial x}\big|_p & \frac{\partial}{\partial y}\big|_p & \frac{\partial}{\partial z}\big|_p \\ f_x & f_y & f_z \end{bmatrix}\right)$$

Given a point $p \in U_3$, suppose we take a very small ball (*not* a point; a ball) and anchor its origin at p.

Informally speaking, the "flow" of the vector field X around p will make the ball start spinning about some axis with some rotational velocity. $\text{curl}_3(X)_p$ points along the axis of rotation with the rotational velocity as its magnitude.

curl_2 takes in a 2-dimensional vector field $X = f_x\frac{\partial}{\partial x} + f_y\frac{\partial}{\partial y} \in \mathcal{X}(U_2)$ and produces a $C^\infty(U_2)$ function,

$$\text{curl}_2(X) \in C^\infty(U_2)$$
$$\text{curl}_2(X) = \det\left(\begin{bmatrix} \frac{\partial}{\partial x}\big|_p & \frac{\partial}{\partial y}\big|_p \\ f_x & f_y \end{bmatrix}\right)$$

A certain physics person once told me that curl is inherently a 3-dimensional concept. In fact, we can use the same idea as above, but we must first "think of" X as a 3-dimensional vector field,

$$X' = f_x\frac{\partial}{\partial x} + f_y\frac{\partial}{\partial y} + 0\frac{\partial}{\partial z}$$

If we take $\text{curl}_3(X')_p$, then it will only have a component along $\hat{k}$, and the coordinate along $\hat{k}$ is the value we give to $\text{curl}_2(X)_p$.

So first we immerse our given vector field in 3d, and then we repeat our ball visualization. Now the axis of rotation of the ball will always be parallel to the Z-axis, so we can get away with just storing the "magnitude" of rotation.

From here onwards, we will just write curl instead of distinguishing between curl_2 and curl_3, leaving it up to the reader to determine it from context.

Divergence (div)

Given an n-dimensional vector field $X = f_{x_1}\frac{\partial}{\partial x_1} + ... + f_{x_n}\frac{\partial}{\partial x_n} \in \mathcal{X}(U_n)$, its "divergence" is a $C^\infty(U)$ function defined as

$$\text{div}(X) \in C^\infty(U)$$
$$\text{div}(X)_p = \frac{\partial f_{x_1}(p)}{\partial x_1} + ... + \frac{\partial f_{x_n}(p)}{\partial x_n}$$

Informally speaking, $\text{div}(X)_p$ measures the extend to which the vector field X acts like a source or sink near the point p.

This may all seem like very disparate concepts, but we will soon see how grad, curl and div are just incarnations of the standard exterior derivative.
